LESSON 01 / 07

A recurrence converts powers into two base cases

For $n \geq 0$, define

$$I_n = \int_0^{\pi/2} \sin^n x \, dx.$$

For $n \geq 2$, integration by parts gives

$$I_n = (n-1)\int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx = (n-1)(I_{n-2} - I_n),$$

so

$$I_n = \tfrac{n-1}{n} I_{n-2}, \qquad I_0 = \tfrac{\pi}{2}, \qquad I_1 = 1.$$

Consequently,

$$I_{2m} = \tfrac{(2m-1)!!}{(2m)!!}\tfrac{\pi}{2}, \qquad I_{2m+1} = \tfrac{(2m)!!}{(2m+1)!!}.$$

WORKED EXAMPLE

An odd-power check, followed by SIS and Q30

For the additional odd-power check $\sin^5 x$,

$$I_5 = \tfrac{4}{5} I_3 = \tfrac{4}{5} \cdot \tfrac{2}{3} I_1 = \tfrac{8}{15}.$$

For SIS and Q30, the requested power is $\sin^6 x$:

$$I_6 = \tfrac{5}{6} \cdot \tfrac{3}{4} \cdot \tfrac{1}{2} \cdot \tfrac{\pi}{2} = \tfrac{5\pi}{32}.$$

Wallis reduction separates even and odd powers

Power or index	Reduction route	Exact value
n=0	base integral of 1	pi/2
n=1	base integral of sin x	1
n=5	(4/5)(2/3) I1	8/15
n=6	(5/6)(3/4)(1/2) I0	5 pi/32
even 2m	double-factorial form times pi/2	decreasing in m
odd 2m+1	double-factorial form	decreasing in m

READ THE FIGURE

The recurrence is exact for every n>=2. It also supplies the ratios needed to prove the infinite Wallis product rather than merely quoting it.

The infinite product follows by squeezing adjacent integrals

On $[0, \pi/2]$, $0 \le \sin x \le 1$, hence

$$I_{2m+1} \le I_{2m} \le I_{2m-1}.$$

The recurrence gives $I_{2m-1}/I_{2m+1} = (2m+1)/(2m) \to 1$, so $I_{2m}/I_{2m+1} \to 1$. Substituting the closed forms and rearranging yields Wallis's product:

$$\frac{\pi}{2} = \lim_{m\to\infty} \frac{2}{1}\frac{2}{3}\frac{4}{3}\frac{4}{5}\cdots\frac{2m}{2m-1}\frac{2m}{2m+1}.$$

This is a convergent infinite product, not an assertion that any finite partial product equals $\pi/2$.

COMMON TRAP

Double factorials skip every other factor

For example, $6!! = 6\cdot 4\cdot 2$ and $5!! = 5\cdot 3\cdot 1$. They do not mean repeated factorial operations. Always retain the base factor $\pi/2$ for even powers and omit it for odd powers.

LESSON 02 / 07

Two convergent parameter integrals and their bridge

For $z>0$, the Gamma function is

$$\Gamma(z)=\int_0^\infty t^{z-1}e^{-t}dt, \qquad \Gamma(z+1)=z\Gamma(z),$$

with $\Gamma(1)=1$ and $\Gamma(n+1)=n!$. For $p,q>0$,

$$B(p,q)=\int_0^1 t^{p-1}(1-t)^{q-1}dt=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

The product identity follows from a double integral and a factorising change of variables. The substitution $t=\sin^2\theta$ gives

$$B(p,q)=2\int_0^{\pi/2}\sin^{2p-1}\theta\cos^{2q-1}\theta\,d\theta.$$

WORKED EXAMPLE

S18 and Q39: the base Beta value

$$B\left(\tfrac12,\tfrac12\right)=2\int_0^{\pi/2}1\,d\theta=\pi=\frac{\Gamma(1/2)^2}{\Gamma(1)}.$$

Since $\Gamma(1/2)>0$, this also gives $\Gamma(1/2)=\sqrt{\pi}$.

The same Beta formula evaluates Q40 and S19/Q43

For $p>-1$, putting $q=1/2$ in the trigonometric Beta form gives

$$\int_0^{\pi/2}\sin^p x\,dx=\tfrac12 B\left(\frac{p+1}{2},\tfrac12\right)=\frac{\sqrt{\pi}}{2}\frac{\Gamma((p+1)/2)}{\Gamma((p+2)/2)}.$$

For Q40, first use $\sin x\cos x=\sin(2x)/2$. With $u=2x$ and sine symmetry on $[0,\pi]$,

$$A=\frac{1}{\sqrt2}\int_0^{\pi/2}\sqrt{\sin(2x)}\,dx=\frac{1}{2\sqrt2}\int_0^{\pi}\sqrt{\sin u}\,du=\frac{1}{\sqrt2}\int_0^{\pi/2}\sqrt{\sin u}\,du.$$

Now put $p=1/2$ in the Beta formula:

$$A=\frac{1}{\sqrt2}\frac{\sqrt{\pi}}{2}\frac{\Gamma(3/4)}{\Gamma(5/4)}\approx 0.8472130848.$$

For S19/Q43, put $p=6$:

$$\int_0^{\pi/2}\sin^6 x\,dx=\frac{\sqrt{\pi}}{2}\frac{\Gamma(7/2)}{\Gamma(4)}=\frac{5\pi}{32},$$

using $\Gamma(7/2)=15\sqrt{\pi}/8$ and $\Gamma(4)=6$.

Q42 and Q44: reflection and duplication

For $0 < z < 1$ in the real-integral setting, Euler's reflection formula is

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Analytic continuation extends it to nonintegral complex values and explains Gamma's poles at the nonpositive integers.

Legendre's duplication formula, for $z > 0$, is

$$\Gamma(z)\Gamma\left(z+\tfrac{1}{2}\right) = 2^{1-2z}\sqrt{\pi}\,\Gamma(2z).$$

At $z = 1/2$, it reduces to $\Gamma(1/2) = \sqrt{\pi}$, not $\pi = 2\sqrt{\pi}$; the source's substitution at that point is corrected. For P18/Q41, $z = 1/4$ gives

$$\Gamma\left(\tfrac{1}{4}\right)\Gamma\left(\tfrac{3}{4}\right) = 2^{1/2}\sqrt{\pi}\,\Gamma\left(\tfrac{1}{2}\right) = \sqrt{2\pi},$$

in agreement with Q42's reflection formula.

Parameter identities must be checked at the substituted value

Source item	Parameter choice	Conclusion
S18 / Q39	p=q=1/2	B=pi and Gamma(1/2)=sqrt(pi)
Q40	p=1/2 after u=2x	0.8472130848 approximately
S19 / Q43	p=6	5 pi/32
Q42 / Q44	reflection and duplication	identities with stated domains
P18 / Q41	z=1/4	product=sqrt(2) pi

READ THE FIGURE

The z=1/2 duplication step is only a consistency check. The z=1/4 evaluation is the nontrivial product and agrees with reflection.

COMMON TRAP

Parameter restrictions are convergence restrictions

The stated real-integral definitions require $z > 0$ and $p, q > 0$. Analytic continuation extends Gamma to other complex values, but it does not make the original real integral converge there.

LESSON 03 / 07

S16/Q32: polar coordinates square a one-dimensional integral

Let

$$I = \int_0^\infty e^{-x^2} dx.$$

The exponential tail proves convergence, and $I > 0$. Therefore

$$I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx\, dy.$$

On the first quadrant, polar coordinates give $x^2 + y^2 = r^2$ and Jacobian r:

$$I^2 = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r\, dr\, d\theta = \tfrac{\pi}{2} \cdot \tfrac{1}{2} = \tfrac{\pi}{4}.$$

Thus

$$\int_0^\infty e^{-x^2} dx = \tfrac{\sqrt{\pi}}{2}, \qquad \int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}.$$

P16/Q33 and Q46: scale, shift and complete the square

For $a > 0$, $u = \sqrt{a}\,x$ gives

$$\int_{-\infty}^\infty e^{-ax^2} dx = \tfrac{1}{\sqrt{a}} \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\tfrac{\pi}{a}}.$$

P16/Q33 is the case $a = 2$, with value $\sqrt{\pi/2}$.

For Q46, also complete the square:

$$-ax^2 + bx = -a\left(x - \tfrac{b}{2a}\right)^2 + \tfrac{b^2}{4a}.$$

Translation does not change a full-line integral, so

$$\int_{-\infty}^\infty e^{-ax^2 + bx} dx = e^{b^2/(4a)} \sqrt{\tfrac{\pi}{a}}, \qquad a > 0.$$

The exponential factor comes from the completed-square constant; the factor $\sqrt{\pi/a}$ comes from scaling.

Additional extension: differentiate a cosine parameter

Define

$$J(a) = \int_0^\infty e^{-x^2} \cos(2ax) dx.$$

Differentiation under the integral is justified by the integrable dominating function $2xe^{-x^2}$. Hence

$$J'(a) = -\int_0^\infty 2xe^{-x^2}\sin(2ax)dx.$$

Integration by parts with $d(e^{-x^2}) = -2xe^{-x^2}dx$ gives

$$J'(a) = -2aJ(a), \qquad J(0) = \frac{\sqrt{\pi}}{2}.$$

Solving yields $J(a) = J(0)e^{-a^2}$. In particular, $J(1) = \sqrt{\pi}/(2e)$, and the full-line cosine transform is $\sqrt{\pi}e^{-a^2}$.

The Gaussian and its transform are positive, exact and checkable

Integral	Method	Value
S16 / Q32 half-line	square and use polar coordinates	sqrt(pi)/2
P16 / Q33 with a=2	scale x by sqrt(a)	sqrt(pi/2)
Q46 with linear term	complete square, translate and scale	sqrt(pi/a) exp(b^2/(4a))
additional cosine transform	differentiate then integrate by parts	sqrt(pi) exp(-a^2)/2

READ THE FIGURE

The three canonical Gaussian results all follow from the same half-line value. The cosine transform is an additional parameter-differentiation extension.

COMMON TRAP

The Jacobian and scaling factors are essential

Replacing $dx\,dy$ by $dr\,d\theta$ without r would give the wrong polar value. Likewise, $u = \sqrt{a}x$ contributes $1/\sqrt{a}$. Also, $I^2 = \pi/4$ determines $|I|$; positivity selects $I = \sqrt{\pi}/2$.

LESSON 04 / 07

Scaling two copies of the same function

Suppose f is continuous with finite limiting values $f(0)$ and $f(\infty)$. For $a, b > 0$, Frullani's formula is

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

Indeed, after $u = ax$ and $u = bx$ in the two pieces of a finite cutoff integral,

$$\int_\varepsilon^R \frac{f(ax) - f(bx)}{x} dx = \int_{a\varepsilon}^{b\varepsilon} \frac{f(u)}{u} du - \int_{aR}^{bR} \frac{f(u)}{u} du.$$

The endpoint limiting values produce the stated logarithm.

WORKED EXAMPLE

S17/Q34 and P17/Q38: exponential Frullani integrals

For $f(t) = e^{-t}$, $f(0) = 1$ and $f(\infty) = 0$. Therefore

$$\int_0^\infty \frac{e^{-x} - e^{-2x}}{x} dx = \ln 2,$$

and

$$\int_0^\infty \frac{e^{-3x} - e^{-5x}}{x} dx = \ln \frac{5}{3} \approx 0.5108256238.$$

More generally, for $x > 0$, $\ln x = \int_0^\infty (e^{-t} - e^{-xt})/t \, dt$.

Additional conditional example: a cosine difference

For $a, b > 0$, $\cos(ax)$ has no limit at infinity, so the basic finite-$f(\infty)$ statement does not directly apply. Introduce $\delta > 0$ and use

$$\frac{\cos(ax) - \cos(bx)}{x} = \int_a^b \sin(tx) dt.$$

Fubini's theorem for the damped integral gives

$$I_\delta = \int_a^b \frac{t}{\delta^2 + t^2} dt = \frac{1}{2} \ln \frac{\delta^2 + b^2}{\delta^2 + a^2}.$$

Dirichlet convergence and Abel continuity as $\delta \to 0^+$ then give

$$\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx = \ln \frac{b}{a}.$$

Source-warning correction: the sine difference converges to zero

For the additional sine-difference example,

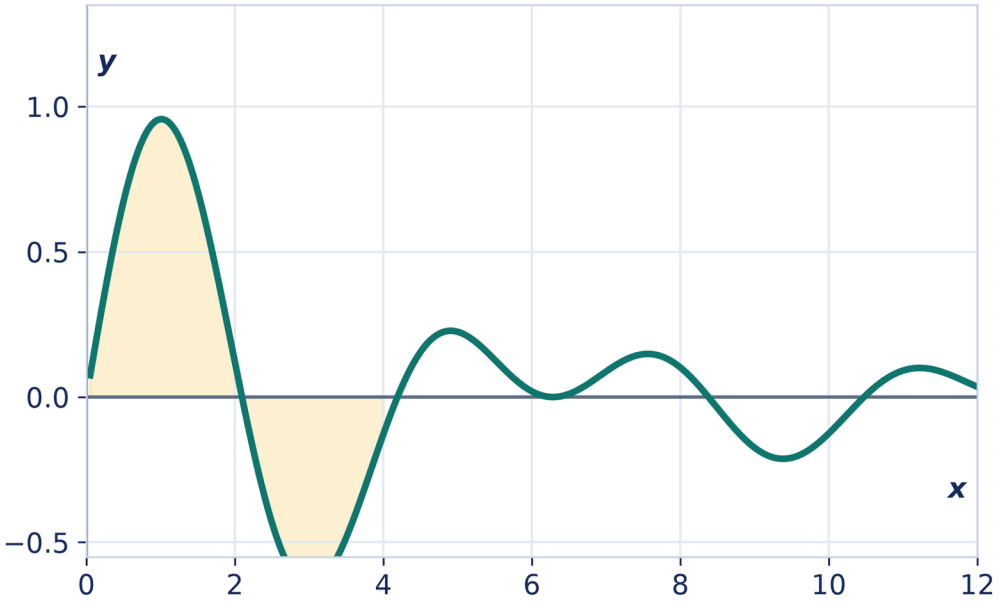
$$\frac{\sin(ax)-\sin(bx)}{x} = -\int_a^b \cos(tx)\,dt.$$

With the same damping,

$$I_\delta = -\int_a^b \frac{\delta}{\delta^2+t^2}\,dt = -\left[\tan^{-1}\frac{t}{\delta}\right]_a^b \longrightarrow 0.$$

Dirichlet's test proves convergence of the original improper integral, and Abel continuity identifies its value as 0. The source's claim that this difference diverges merely because sine has no limiting value at infinity is incorrect.

A cosine Frullani integrand with a1 and b2



READ THE FIGURE
For a=1 and b=2, the full conditional integral is ln(2). The shaded finite window is illustrative; convergence and the exact value come from the damped calculation, not the plot.

COMMON TRAP
No f(infinity) does not automatically mean divergence
The finite-limit Frullani theorem has hypotheses, but failure of one shortcut does not settle convergence. The exponential source questions satisfy the hypotheses directly; oscillatory differences require a separate convergence argument and can converge conditionally.

LESSON 05 / 07

Differentiate a parameter, then recover the constant

Under suitable local domination and convergence hypotheses,

$$I(a)=\int f(x,a)\,dx \quad\Longrightarrow\quad I'(a)=\int \frac{\partial f}{\partial a}(x,a)\,dx.$$

After evaluating the simpler derivative, integrate in a and use one known value to determine the constant.

WORKED EXAMPLE

S14/Q28 and P14/Q29: differentiate a power integral

For $a>-1$,

$$F(a)=\int_0^1 x^a\,dx=\frac{1}{a+1}.$$

Differentiating once and twice gives

$$\int_0^1 x^a\ln x\,dx=-\frac{1}{(a+1)^2},\qquad \int_0^1 x^a(\ln x)^2\,dx=\frac{2}{(a+1)^3}.$$

At $a=1$, the first identity gives S14/Q28 $=-1/4$. At $a=0$, the second gives P14/Q29 $=2$. These agree with the by-parts evaluations in Lesson 8.

Q45: a logarithmic quotient represented by a parameter integral

For $a>-1$, define

$$I(a)=\int_0^1 \frac{x^a-1}{\ln x}\,dx.$$

At $x=1$, the expression has continuous limiting value a; convergence at 0 follows from the x^a scale and the logarithmic denominator. Differentiation gives

$$I'(a)=\int_0^1 x^a\,dx=\frac{1}{a+1}.$$

Since $I(0)=0$,

$$I(a)=\int_0^a \frac{dt}{1+t}=\ln(1+a).$$

Thus Q45, with $a=3$, equals $\ln 4=2\ln 2$.

The two anchors are consistent: $I(0)=0=\ln 1$, and $I'(a)=1/(1+a)$ is exactly the derivative of $\ln(1+a)$. For $-1<a<0$, the oriented parameter integral and the logarithm are both negative, providing an additional sign check.

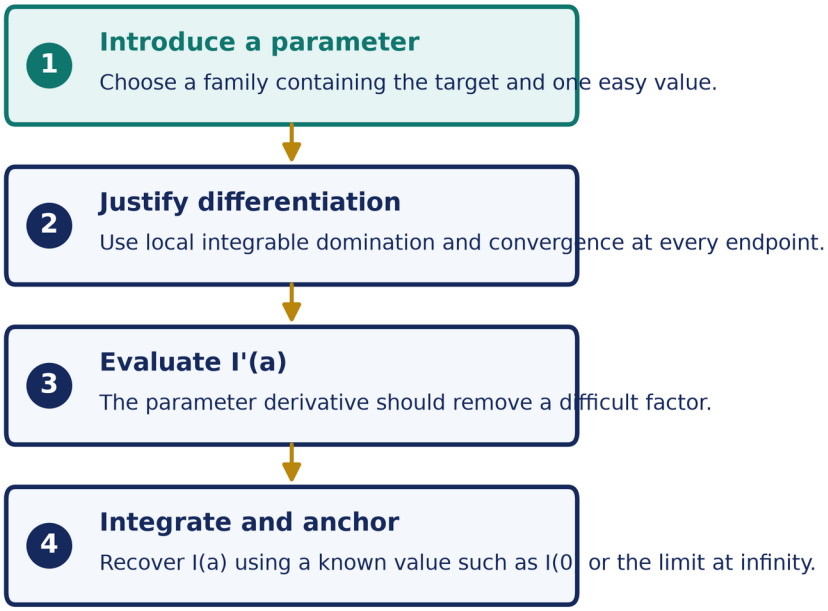
Q45's source representation needs a corrected lower limit

For $x > 0$, $x \neq 1$,

$$\int_0^a x^t dt = \left[\frac{x^t}{\ln x}\right]_0^a = \frac{x^a - 1}{\ln x}.$$

If the lower limit were 1 as printed in the source, the value would instead be $(x^a - x)/\ln x$. Thus the intended representation requires lower limit 0; the final Q45 answer $\ln 4$ remains correct.

Feynman differentiation is a four-step method with hypotheses



READ THE FIGURE
For Q45, the anchor is $I(0)=0$. For the power derivatives, differentiation gives the answer directly without an integration constant.

COMMON TRAP
Differentiation under the integral sign needs its own hypotheses
Convergence of the original integral alone is not sufficient. Use local integrable domination for the parameter derivative, check improper endpoints, and anchor the integrated result with a known value.

LESSON 06 / 07

First evaluate the damped parameter family

For $t > 0$, define

$$J(t) = \int_0^\infty e^{-tx}\frac{\sin x}{x}dx,$$

with continuous value 1 for $\sin x/x$ at $x = 0$. Local domination permits differentiation:

$$J'(t) = -\int_0^\infty e^{-tx}\sin x\,dx = -\frac{1}{1+t^2}.$$

Since $J(t) \to 0$ as $t \to \infty$,

$$J(t) = \int_t^\infty \frac{ds}{1+s^2} = \frac{\pi}{2} - \tan^{-1} t.$$

This proves the damped formula without setting $t = 0$ prematurely.

Convergence comes from oscillation and a decreasing envelope

The expression $\sin x/x$ has continuous extension 1 at $x = 0$, so only infinity is improper. For $1 < R < S$, integration by parts gives

$$\int_R^S \frac{\sin x}{x}dx = \left[-\frac{\cos x}{x}\right]_R^S - \int_R^S \frac{\cos x}{x^2}dx.$$

The boundary terms are $O(1/R)$ and the remaining absolute integral is at most $\int_R^\infty x^{-2}dx = 1/R$. Thus the finite integrals have a limit.

The convergence is not absolute. On each interval $[n\pi + \pi/6, n\pi + 5\pi/6]$, $|\sin x| \geq 1/2$, so the contribution to $\int |\sin x|/x\,dx$ is at least a constant multiple of $1/(n+1)$. The harmonic series diverges.

Abel continuity supplies the exact value

For $\delta > 0$, the calculation above gives

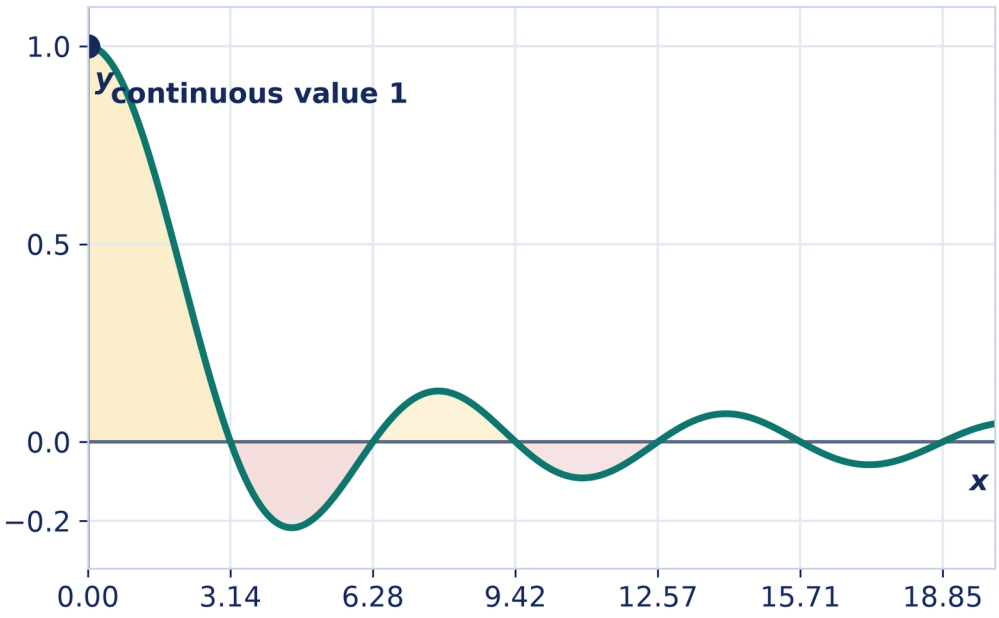
$$I(\delta) = \int_0^\infty e^{-\delta x}\frac{\sin x}{x}dx = \frac{\pi}{2} - \tan^{-1}\delta.$$

Because the undamped improper integral converges, Abel continuity gives Q37:

$$\int_0^\infty \frac{\sin x}{x}dx = \lim_{\delta \to 0^+} I(\delta) = \frac{\pi}{2}.$$

This is the missing justification for inserting a convergence factor and then removing it.

Alternating Dirichlet arches have a finite signed sum



READ THE FIGURE

Alternating arches make the improper integral converge conditionally to pi/2. Their absolute areas do not have a finite sum, and the graph cannot by itself establish the exact limit.

COMMON TRAP

Conditional convergence is not absolute convergence

The exact value $\pi/2$ concerns the ordered improper integral $\lim_{R\to\infty}\int_0^R \sin x/x\,dx$. It does not assert that $\int_0^\infty |\sin x/x|dx$ is finite, and it does not permit arbitrary rearrangement of positive and negative area.

LESSON 07 / 07

Two rational tails can be evaluated without numerical approximation

For the first additional rational tail, partial fractions give

$$\frac{1}{(x^2+1)(x^2+4)} = \frac{1}{3}\left(\frac{1}{x^2+1} - \frac{1}{x^2+4}\right).$$

Therefore

$$\int_0^\infty \frac{dx}{(x^2+1)(x^2+4)} = \frac{1}{3}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{\pi}{12}.$$

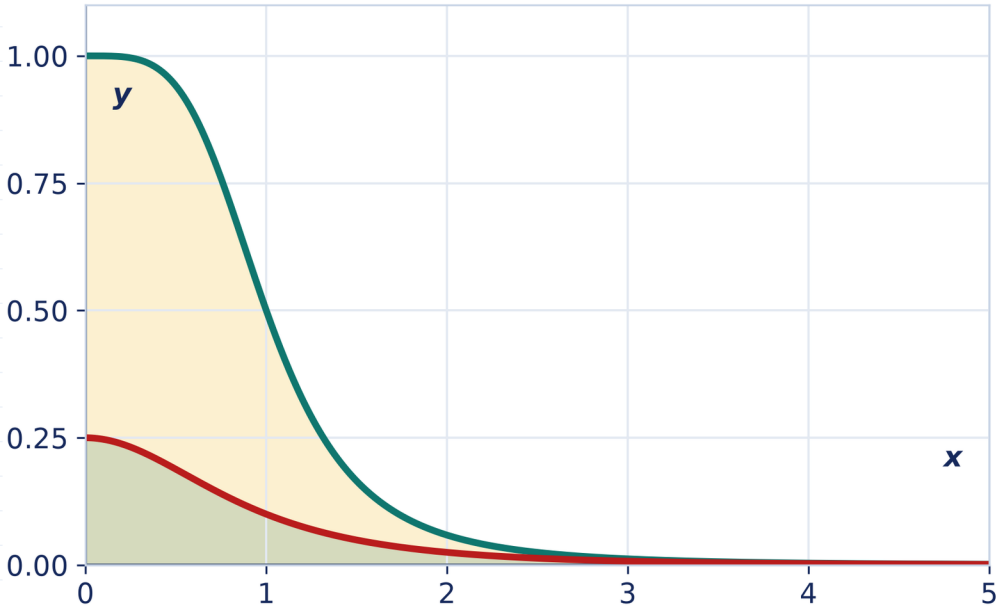
For the additional quartic tail, let $I = \int_0^\infty dx/(1+x^4)$. Inversion $x \mapsto 1/x$ gives $I = \int_0^\infty x^2 dx/(1+x^4)$. Hence

$$2I = \int_0^\infty \frac{1+x^2}{1+x^4}dx.$$

Divide by x^2 and put $u = x - 1/x$, so $du = (1+1/x^2)dx$ and u runs from $-\infty$ to ∞:

$$2I = \int_{-\infty}^{\infty} \frac{du}{u^2+2} = \frac{\pi}{\sqrt{2}}, \qquad I = \frac{\pi}{2\sqrt{2}}.$$

Two positive rational tails with exact areas



READ THE FIGURE
The teal curve 1/(1+x^4) has full tail area pi/(2 sqrt(2)); the red curve has full tail area pi/12. Only [0,5] is plotted, while the proofs include the remaining tails.

Log-sine uses reflection and a doubling substitution

For the additional log-sine integral, set

$$I=\int_0^{\pi/2}\ln\sin x\,dx.$$

The endpoint singularities are integrable. Reflection gives $I=\int_0^{\pi/2}\ln\cos x\,dx$, so

$$2I=\int_0^{\pi/2}\ln(\sin x\cos x)dx=\int_0^{\pi/2}\ln\sin(2x)dx-\tfrac{\pi}{2}\ln 2.$$

With $u=2x$ and symmetry on $[0,\pi]$,

$$\int_0^{\pi/2}\ln\sin(2x)dx=\tfrac{1}{2}\int_0^{\pi}\ln\sin u\,du=I.$$

Thus $2I=I-(\pi/2)\ln 2$, and

$$\int_0^{\pi/2}\ln\sin x\,dx=-\tfrac{\pi}{2}\ln 2.$$

A log-tangent consequence follows immediately:

$$\int_0^{\pi/2}\ln\tan x\,dx=\int_0^{\pi/2}\ln\sin x\,dx-\int_0^{\pi/2}\ln\cos x\,dx=0.$$

COMMON TRAP

Logarithmic identities require positive inputs inside each interval

Use $\ln(\sin x\cos x)$ on $(0,\pi/2)$, where both factors are positive, and interpret endpoints through limits. The identity $\ln(ab)=\ln a+\ln b$ is not applied at an endpoint where either logarithm is undefined.